

SMTE – GutBrainIE@CLEF 2026: Biomedical Named Entity Recognition and Mention-Level Relation Extraction for the Gut-Brain Axis

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Abstract

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Biomedical NLP, Named Entity Recognition, Relation Extraction, Gut-Brain Axis, Transformer Models, CLEF BioASQ

1. Introduction

The gut-brain axis — the bidirectional communication network between the gastrointestinal tract and the central nervous system — has recently emerged as a major research focus in biomedicine. A growing body of studies highlights the role of gut microbiota dysbiosis in neurological and psychiatric conditions, including Alzheimer’s disease, Parkinson’s disease, Multiple Sclerosis, Amyotrophic Lateral Sclerosis, and various mental health disorders.

However, the rapid increase in biomedical publications makes manual analysis and curation infeasible, creating the need for automatic Information Extraction (IE) systems capable of structuring this knowledge.

The GutBrainIE shared task at CLEF 2026 [1], part of the BioASQ Lab and developed within the EU-supported HEREDITARY project, addresses this challenge by focusing on structured information extraction from PubMed abstracts related to the gut-brain axis.

Participants are required to extract both entity mentions and semantic relations. In this work, we participate in two subtasks:

- **Subtask 6.1.1 – Named Entity Recognition (NER):** identification and classification of entity mentions into 13 predefined biomedical categories;
- **Subtask 6.2.1 – Mention-Level Relation Extraction (M-RE):** identification of semantic relations between entity mentions, expressed as subject–predicate–object triples.

Both subtasks present significant challenges. In NER, several categories are semantically close (e.g., chemical, drug, and dietary supplement), making classification difficult even for domain experts. Additionally, the dataset is highly imbalanced, with frequent categories such as human and microbiome dominating rare ones like food or gene.

In M-RE, the main challenges include the large number of possible entity pairs, the sparsity of positive relations, and the propagation of errors from the NER stage.

Another key difficulty lies in the heterogeneity of the training data. The dataset combines annotations of different quality levels — Gold (expert), Silver (student), Silver 2025 (partially automatic), and Bronze (fully automatic) — requiring careful strategies to effectively leverage all available data.

In this paper, we present our system, SMTE, which addresses both subtasks using transformer-based models fine-tuned on the GutBrainIE dataset. For NER, we explore multiple training strategies,

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including priority-based merging, curriculum learning, and annotator-weighted training, and combine the strongest models into a hybrid ensemble. For M-RE, we adopt a pairwise classification approach with entity markers and type-based candidate filtering.

The remainder of the paper is organized as follows. Section 2 describes the dataset and task setup. Section 3 presents the NER system. Section 4 describes the relation extraction approach. Section 5 concludes the paper.

2. Task and Data Description

The GutBrainIE shared task [1] focuses on extracting structured information from PubMed abstracts describing the interaction between gut microbiota and neurological or mental health conditions.

2.1. Subtask 6.1.1 – Named Entity Recognition

In this subtask, systems identify and classify entity mentions into 13 predefined biomedical categories: anatomical location, animal, bacteria, biomedical technique, chemical, DDF (disease, disorder, or finding), dietary supplement, drug, food, gene, human, microbiome, and statistical technique.

Each entity mention is represented as a tuple:

$$(entityCategory; entityLocation; startOffset; endOffset)$$

where the location specifies whether the entity appears in the title or the abstract.

2.2. Subtask 6.2.1 – Mention-Level Relation Extraction

In this subtask, systems identify semantic relations between entity mentions within the same document.

Each relation is expressed as a triple:

$$(subjectMention; relationPredicate; objectMention)$$

The relation schema defines directed predicates between specific pairs of entity types. Examples include:

- bacteria → DDF: influence
- chemical → microbiome: impact, produced by
- microbiome → DDF: is linked to
- DDF → bacteria: change abundance

2.3. Dataset

The training data is composed of four collections with different levels of annotation quality:

- **Gold:** high-quality annotations created by expert annotators;
- **Silver:** annotations produced by trained students, divided into two groups based on annotation consistency;
- **Silver 2025:** annotations from a previous edition, partially enriched with automatically generated concept-level information;
- **Bronze:** distantly supervised annotations generated automatically without manual validation.

The test set consists of a held-out subset of the Gold data, ensuring high-quality evaluation.

3. Named Entity Recognition

We frame NER as a token classification task over BIO-tagged sequences. All 13 entity categories are expanded into a BIO label set of 27 tags (one B-/I- pair per class, plus O). Each document is split into two independent segments — title and abstract — so that character offsets remain consistent with the annotated location field.

3.1. System Overview

Our final submitted system is a **hybrid ensemble** combining two complementary transformer models: BioBERT fine-tuned on Gold + Silver + Bronze data (NB1b), and PubMedBERT fine-tuned on the same data (NB1b_{pubmed}). The ensemble achieves Macro-F1 0.802 and Micro-F1 0.8324 on the development set, outperforming every individual model.

The design of the ensemble was informed by a systematic comparison of seven training configurations (NB1–NB4b), all sharing a common inference pipeline. The following subsections describe training strategies, the unified inference procedure, and ensemble construction in detail.

3.2. Training Strategy

Backbone models and shared hyperparameters. We use transformer-based biomedical encoders for all NER experiments. Our main backbone is BioBERT [2] (dmis-lab/biobert-v1.1), and we also evaluate a variant based on PubMedBERT [3] (microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract). In all settings, the encoder is fine-tuned end-to-end with a token-level classification head.

Unless otherwise stated, all configurations use the same training setup:

- learning rate 3×10^{-5} with linear scheduling and 10% warmup;
- weight decay 0.01;
- batch size 8 with gradient accumulation over 2 steps;
- best checkpoint selected on the development set;
- class-weighted cross-entropy loss, where class weights are computed from inverse label frequency^{0.5} and clipped to the range [0.5, 5.0].

Experimental settings. All model variants were first developed and compared on the development set. We explored several strategies for combining training data of different quality.

- **NB1 – Single-stage training on Gold + Silver (BioBERT).** In this setting, Gold and Silver data are merged into a single training set using a priority-based policy. When the same PMID is available in more than one source, the annotation from the higher-quality source is kept. The model is then trained in a single stage for 5 epochs.
- **NB1b – Single-stage training on Gold + Silver + Bronze (BioBERT / PubMedBERT).** This configuration extends NB1 by also including Bronze data after the same priority merge. We test this setup with two different backbones: BioBERT and PubMedBERT. This is the strongest single-stage setting, and it allows the model to benefit from a larger and more diverse training set while still giving priority to higher-quality annotations.
- **NB2 – Two-stage curriculum training (BioBERT).** In this variant, training is divided into two sequential stages. The first stage uses Gold data only, while the second stage continues training from the first checkpoint on a mixture of oversampled Gold and shuffled Silver data. The goal is to expose the model to cleaner supervision first and then introduce noisier annotations in a controlled way.
- **NB3 – Three-stage cross-year curriculum (BioBERT).** This setting adds the 2025 collection as an initial warm-up stage. Training starts from 2025 Platinum + Gold data, then continues on

2026 Gold, and finally on a 2026 Gold + Silver mixture. The motivation is to exploit additional domain-relevant supervision, although differences between the 2025 and 2026 annotations may introduce extra noise.

- **NB4 / NB4b – Annotator-weighted training (BioBERT).** Instead of merging data with a hard priority rule, these variants assign different sample weights according to the annotation source. Expert annotations receive the highest weight, followed by Student A, Student B, Silver 2025, and Bronze. NB4 uses Gold + Silver + Silver 2025, while NB4b also includes Bronze. This strategy aims to keep all available training instances while reducing the effect of noisier annotations.

Selected models for the final system. Among the single-model configurations, NB1b gives the strongest overall results, especially when using the PubMedBERT backbone. Based on these experiments, we keep both NB1b-BioBERT and NB1b-PubMedBERT as the main NER models. The final submitted NER run is obtained through a post-hoc hybrid ensemble that combines the predictions of the two models. The ensemble strategy is described in Section 3.4.

3.3. Inference

All models are evaluated with the same inference pipeline, applied post-hoc to every checkpoint. No model-specific retraining is performed at inference time, so the observed differences depend only on training.

Step 1 – BIO decoding and confidence scoring. Each title or abstract is tokenised with offset mapping enabled. The predicted BIO sequence is decoded into entity spans, and each span is assigned a confidence score computed as the median token probability within the span:

$$\text{score}(e) = \text{median}_{t \in e} p(y_t \mid \mathbf{x})$$

Using the median makes the score less sensitive to low-confidence boundary tokens.

Step 2 – Label-specific thresholding. We apply a two-pass thresholding strategy. In the first pass, entities are kept only if their confidence exceeds a high threshold specific to the predicted label. In the second pass, lower thresholds are used only for a small set of recall-critical labels (chemical, bacteria, food, and dietary supplement). Entities recovered in the second pass are added only when they do not conflict with spans already selected in the first pass. This design improves recall for difficult labels while keeping precision under control.

The label-specific rules were designed from development-set error analysis and target the most frequent boundary and label-confusion errors observed in preliminary runs.

Step 3 – Label-aware post-processing. We then apply a small set of deterministic label-specific heuristics. First, obvious false positives are removed, including very short spans, markup fragments, and overly generic terms for selected labels. Second, systematic confusions between semantically close categories are corrected with surface-form rules. In particular, gene-like mentions initially predicted as chemical are remapped to gene, while some chemical-like mentions predicted as gene are remapped in the opposite direction. Similarly, food-related and supplement-related surface patterns are used to reduce confusion between food and dietary supplement.

Step 4 – Deduplication and overlap handling. Finally, duplicate predictions are removed and overlapping spans are resolved with the same logic used by the official evaluator, ensuring consistency between development experiments and final submission files.

3.4. Ensemble

The final NER system combines the predictions of two models: NB1b-BioBERT and NB1b-PubMedBERT. The ensemble is applied as a post-processing step, without additional training.

Motivation. Although PubMedBERT achieves the best overall performance as a single model, the two models show complementary behavior across entity types. In particular, BioBERT recovers additional correct mentions for some labels where PubMedBERT is more conservative. Combining their predictions therefore improves overall coverage while maintaining precision.

Ensemble strategy. We evaluate several merging strategies on the development set, including union, intersection, and hybrid approaches. The best results are obtained with a simple hybrid strategy that uses PubMedBERT as the base model and adds non-overlapping predictions from BioBERT.

Merge procedure. The final merge is performed as follows:

1. All spans predicted by PubMedBERT are kept as the base set.
2. Spans predicted by BioBERT are added only if they do not overlap with any span in the base set.
3. The resulting set is deduplicated based on character offsets.
4. Overlapping spans are resolved by keeping the longer span.

This procedure improves recall by incorporating additional valid spans from BioBERT, while preserving the higher precision of PubMedBERT.

3.5. Experiments and Results

Evaluation metrics. We follow the official evaluation protocol, which requires exact span match: a prediction is correct only if both label and character offsets match the gold annotation. We report Macro-F1 and Micro-F1, along with Precision and Recall.

Results. Table 1 reports the results of all configurations on the development set. Among single models, NB1b_{pubmed} (PubMedBERT) achieves the best performance. The final ensemble further improves the results, reaching Macro-F1 = 0.802 and Micro-F1 = 0.8324, outperforming all individual models.

Table 1

NER results on the development set.

Model	Mac-P	Mac-R	Mac-F1	Mic-P	Mic-R	Mic-F1
NB1	0.8264	0.7215	0.7648	0.8638	0.7572	0.8070
NB1b	0.8091	0.7587	0.7749	0.8480	0.7925	0.8194
NB1b _{pubmed}	0.8305	0.7728	0.7973	0.8693	0.7886	0.8270
NB2	0.8117	0.7307	0.7638	0.8496	0.7616	0.8032
NB3	0.8000	0.6910	0.7381	0.8546	0.7322	0.7887
NB4	0.8027	0.7130	0.7506	0.8548	0.7449	0.7961
NB4b	0.8202	0.7352	0.7685	0.8645	0.7644	0.8114
Ensemble	0.804	0.8068	0.802	0.8403	0.8247	0.8324

Effect of training data. Adding Bronze annotations (NB1b vs. NB1) improves recall and overall performance, showing that noisy data can still provide useful signal. Curriculum-based strategies (NB2, NB3) do not outperform single-stage training, likely due to distribution differences across datasets. Annotator-based weighting (NB4) increases precision but reduces recall.

Effect of backbone. Replacing BioBERT with PubMedBERT yields the largest improvement among single models, confirming the benefit of domain-specific pre-training on biomedical text.

Ensemble. The ensemble combines PubMedBERT predictions with non-overlapping spans from BioBERT. This simple strategy improves recall while preserving precision, consistently outperforming both individual models.

Per-label analysis. Table 2 reports per-label F1 scores. Frequent labels such as *DDF*, *human*, and *microbiome* achieve high performance, while more challenging categories such as *food* and *gene* remain harder due to data scarcity and label ambiguity.

Table 2

Per-label F1 on the development set for selected configurations.

Label	NB1	NB1b	NB1b _{pubmed}	Ensemble
DDF	0.872	0.893	0.878	0.893
anatomical location	0.768	0.758	0.832	0.832
animal	0.852	0.852	0.823	0.864
bacteria	0.715	0.748	0.742	0.748
biomedical technique	0.672	0.698	0.700	0.700
chemical	0.696	0.722	0.738	0.738
dietary supplement	0.741	0.767	0.786	0.786
drug	0.857	0.855	0.850	0.868
food	0.591	0.614	0.746	0.746
gene	0.627	0.606	0.652	0.652
human	0.917	0.904	0.923	0.938
microbiome	0.921	0.910	0.919	0.934
statistical technique	0.714	0.747	0.778	0.778

Official test-set results. After selecting the final configuration on the development set, we generated predictions for the official hidden test set. Table X reports the final submission results returned by the shared-task organizers.

4. Mention-Level Relation Extraction

5. Conclusion

Acknowledgments

TODO: acknowledgments to be added.

Declaration on Generative AI

During the preparation of this work, generative AI tools were used to support code development and writing refinement. All outputs were reviewed and validated by the authors.

References

- [1] HEREDITARY Project, Gutbrainie: Information extraction on gut-brain axis biomedical texts, <https://hereditary.dei.unipd.it/challenges/gutbrainie/2026/>, 2026. Accessed: 2026.

- [2] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, *Bioinformatics* 36 (2020) 1234–1240. doi:10.1093/bioinformatics/btz682.
- [3] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, H. Poon, Domain-specific language model pretraining for biomedical natural language processing, *ACM Transactions on Computing for Healthcare* 3 (2021) 1–23. doi:10.1145/3458754.